

Cong Yang

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Education

2023–2026	Ph.D. in Computer Science • School of Computer Science, Wuhan University • Advisor: Lefei Zhang (zhanglefei@whu.edu.cn) • GPA: 3.71/4.0	Wuhan, China
2020–2023	M.S. in Geological Resources and Geological Engineering • School of Geophysics and Geomatics, China University of Geosciences • Advisor: Yanni Dong (dongyanni@whu.edu.cn) • GPA: 3.62/4.0	Wuhan, China
2016–2020	B.S. in Earth Information Science and Technology • School of Geophysics and Geomatics, China University of Geosciences • GPA: 3.35/4.0	Wuhan, China

Research Interests

Multimodal Learning, Image Captioning, Post-Training

Research Experience

2026–Present	Associative and Diversity-Aware Reinforcement Learning for Multimodal Reasoning <i>Doctoral Researcher, Advisor – Prof. Lefei Zhang</i> • Built the Associative CoT-9K dataset by restructuring linear chains of thought into dependency-linked question–answer subproblem chains, enabling structured and traceable multimodal reasoning. • Implemented a two-stage post-training pipeline comprising supervised fine-tuning on Associative CoT-9K followed by group relative policy optimization (GRPO). • Introduced a semantic-diversity regularizer using Qwen3-Embedding-4B to penalize homogeneous reasoning trajectories and extend optimization from final-answer accuracy to reasoning-path quality. • Evaluated Qwen2.5-VL-7B and Qwen3-VL-8B across multimodal reasoning benchmarks, developing end-to-end experience in reinforcement-learning post-training, ablation design, trajectory analysis, training-cost profiling, and first-author manuscript preparation (M4).
2025–2026	Reliable and Efficient Perception in Vision–Language and Long-Video Models <i>Doctoral Researcher, Advisor – Prof. Lefei Zhang</i> • Built Focus and Awaken, a training-free decoding framework that curbs vision–language hallucination by restoring 2D spatial locality with a Chebyshev attention penalty and routing <0.1% of feed-forward neurons into single-pass dual-track contrastive decoding, cutting sentence- and instance-level CHAIR rates to 46.0 and 13.2 at $1.04\times$ latency (M3).

2023–2026	- Developed UniEP for long-video understanding, where one question-conditioned segment-level evidence plan drives both exact-budget frame selection and hierarchical token pruning, gaining 2.65 points over uniform 32-frame sampling and preserving 99.3% of full-token performance at 10% visual-token retention (M2). Vision–Language Learning for Fine-Grained Image Captioning and Change Understanding <i>Doctoral Researcher, Advisor – Prof. Lefei Zhang</i> - Introduced BITA, a two-stage framework pairing multi-scale frequency features and contrastive learning with prefix causal language modeling for image–text alignment (first-author, J4). - Designed MGIMM and KCFI for fine-grained description and change captioning, combining regional–global feature fusion and key-change perception under multi-task optimization, with 61,783 attribute-guided instruction pairs curated for MGIMM (first-author, J5, J6). - Replaced visual-space differencing with semantically aligned change slots, where learnable queries compete for bi-temporal visual tokens and are aligned to caption semantics via bidirectional soft-Chamfer matching and InfoNCE, reaching the best reported S^* on LEVIR-CC (81.59) and WHU-CC (90.02) (M1).
2021–2022	Metric Learning and Domain Adaptation under Limited Annotations <i>Graduate Researcher, Advisor – Prof. Yanni Dong</i> - Developed DMLOHM, combining discriminative metric-space learning with online hard-triplet mining to improve representation learning under limited annotations. - Achieved gains of 0.13%–4.03% with only 85 labeled samples per class, building experience in sparse-label experimental design, metric-learning evaluation, and difficult-sample analysis. - Designed ADA-DDA to assign attention-based sample weights and dynamically balance marginal and conditional distribution alignment across domains, improving accuracy by 5.36% over the next-best method. - Applied these methods to hyperspectral and cross-domain remote sensing classification, contributing to J1, J2, J3, and a granted Chinese invention patent (CN114202697B).

Publication

Journal Articles

- 2025 J6 **Enhancing Perception of Key Changes in Remote Sensing Image Change Captioning**
Cong Yang, Zuchao Li, Hongzan Jiao, Zhi Gao, Lefei Zhang
IEEE Transactions on Image Processing
DOI: [10.1109/TIP.2025.3589096](https://doi.org/10.1109/TIP.2025.3589096)
- 2024 J5 **MGIMM: Multi-Granularity Instruction Multimodal Model for Attribute-Guided Remote Sensing Image Detailed Description**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2024.3497976](https://doi.org/10.1109/TGRS.2024.3497976)
- J4 **Bootstrapping Interactive Image–Text Alignment for Remote Sensing Image Captioning**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2024.3359316](https://doi.org/10.1109/TGRS.2024.3359316)
- 2022 J3 **Attention-Based Dynamic Alignment and Dynamic Distribution Adaptation for Remote Sensing Cross-Domain Scene Classification**

Cong Yang, Yanni Dong, Bo Du, Liangpei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2022.3225589](https://doi.org/10.1109/TGRS.2022.3225589)

J2 **Joint Distance Transfer Metric Learning for Remote-Sensing Image Classification**
Yanni Dong, Tianyang Liang, **Cong Yang**, Hui Luo, Yuxiang Zhang
IEEE Geoscience and Remote Sensing Letters
DOI: [10.1109/LGRS.2022.3149141](https://doi.org/10.1109/LGRS.2022.3149141)

2021 J1 **Deep Metric Learning with Online Hard Mining for Hyperspectral Classification**
Yanni Dong, **Cong Yang**, Yuxiang Zhang
Remote Sensing
DOI: [10.3390/rs13071368](https://doi.org/10.3390/rs13071368)

Preprints

2026 M4 **AD-GRPO: Associative Diversity-Aware GRPO for Multimodal Reasoning**
Cong Yang, Yuchun Miao, Kangzhe Yuan, Zuchao Li, Lefei Zhang
Under Review

2026 M3 **Focus and Awaken: Mitigating VLM Hallucinations via Chebyshev Attention and Neuron-Routed Contrastive Decoding**
Ziyang Ma, **Cong Yang**, Baoyuan Qi, Haojun Ai, Ping Wang, Zuchao Li
Under Review

2026 M2 **UniEP: Unified Evidence Planning for Efficient Long Video Understanding**
Yuxuan Rao, **Cong Yang**, Kangzhe Yuan, Zuchao Li, Lefei Zhang
Under Review

2026 M1 **Semantically Aligned Change Slots for Remote Sensing Image Change Captioning**
Yuanzhi Yao, **Cong Yang**, Liang Lv, Jun Wan, Lefei Zhang
Under Review

Service

Peer Reviewing

Journals

- *IEEE Transactions on Geoscience and Remote Sensing* (TGRS)
- *IEEE Transactions on Image Processing* (TIP)
- *IEEE Transactions on Circuits and Systems for Video Technology* (TCSVT)

Awards

2023 Outstanding Graduate of China University of Geosciences

2021 China National Scholarship

Skills

Research Expertise

Remote sensing image understanding and generation; Multimodal learning; Model post-training

Programming & Computing

Python, MATLAB, C/C++; PyTorch, Hugging Face Transformers, DeepSpeed; Slurm, distributed multi-GPU computing, Git

Languages & Writing

Mandarin Chinese (native); English (professional working proficiency; experienced in academic writing and manuscript preparation)

Referees

Dr. Lefei Zhang

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Dr. Zuchao Li

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Dr. Yanni Dong

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